<u>Design Report</u>

Fuselage Design

The fuselage underwent many changes from initial design concepts to final design. We streamlined the shape significantly to reduce form drag and improve aerodynamic efficiency.

- **Geometry & Sizing**: The fuselage length was optimised to exactly 284.2mm. This specific dimension was derived using the ideal fuselage length formula $L = \sqrt{4 \times Projected\, Wing\, Area}$ as defined by Zaic (1944). We calculated our wing's projected area ($Projected\, Wing\, Area = 250mm \times cos(10) \times 82mm$). This length minimises the structural weight, reducing our required lift, whilst ensuring an adequate tail moment arm.
- **Wing Positioning**: We positioned the wing such that the leading edge of the wings sits at 25% of the total fuselage length away from the nose of the aircraft as this allows a strong moment arm for balancing. (Zaic, F, 1944).
- **Structure**: We adopted a "sandwich" structure consisting of three layers. The middle layer is made of balsa and has cutouts in strategic areas to save weight and influence the CG. The outer two layers are plywood to provide structural strength as the plywood is stronger and more likely to withstand the impact.
- **Surface Finish**: The wooden fuselage components are sanded with low-grit sandpaper. This does not remove significant material but ensures a smooth airflow, further reducing skin friction drag.

Wing Design

The wing design was driven by the need for high lift and lateral stability in a fixed circular path.

- **Aerofoil & Geometry**: We selected the NACA 2412 for its high lift-to-drag ratio. Based on research into RC planes, we chose an aspect ratio of 6, resulting in a wing span of 500mm and a chord of 82mm.
- **Incidence**: While standard model gliders often use a 2-degree incidence, we increased this to 4 degrees. This higher angle is intended to provide greater lift at low speeds and reduce the risk of stall during tethered flight.
- **Stability**: A dihedral angle of 10 degrees was chosen to provide lateral stability against the continuous sideslip caused by the circular motion. We opted for a high-wing configuration, utilising the "pendulum effect", as described by Simons (1999), to further increase our lateral stability and avoid Dutch roll (oscillations in the yaw and roll direction).

Wing Structure

- The wing is composed of 8 ribs. To balance weight and durability, the standard ribs are laser-cut balsa. However, the wing root (which bears the wing's weight) and the wing tips (which are at high risk of ground impact) are 3D printed for enhanced strength as the plastic will be stronger than the balsa, making the glider more durable in the event of a crash.

- Structural rigidity is provided by a 4mm carbon fiber tube acting as the wing spar. This tube runs spanwise through all 8 ribs, distributing aerodynamic loads across the wing. The carbon fibre was used as it is very rigid and already in a tube shape. This spar serves a dual purpose as the tether attachment point. The tether line is threaded directly through the hollow carbon fiber tube. This design choice ensures the tether pulls from the aircraft's center of lift, enhancing stability during circular flight, guided by the principles of circular flight dynamics (Hepperle, 2003).

Tailplane Design

- We considered various designs for our empennage. We initially considered a T-tail (Torode, H, 2010) as they allow for a better glide ratio and they provide more predictable aerodynamic performance (Bruce, K, 2019). However, in the end, we chose to go with a conventional design for several reasons.

- Firstly, the conventional design is lighter than the T-tail. T-tails require a thicker, heavier vertical stabiliser to be able to support the weight of the horizontal stabiliser above it (Scholz, D, 2015). With the vertical stabiliser being a light-weight, thin piece of balsa, we decided it would not be able to support the horizontal stabiliser. The vertical stabiliser is now a part of the fuselage that will be laser cut rather than a separate piece to make it stronger in case of a crash, as they were quick to fall off previously and also to reduce interference drag as there is no join between 2 components, it is all just one.

- Secondly, a conventional rail eliminates the risk of deep stall (Nguyen, D…, 2022).In a T-tail configuration, the high-mounted tailplane can become immersed in the turbulent wake of the main wings during high angles of attack, causing a loss of lift and potentially a stall (SKYbrary, 2026). Finally, our design reduces material usage by cutting parts out of the end of the fuselage at the rear and offers stronger structural integrity, allowing the empennage to be secured more tightly than a T-tail in the event of a crash.

Motor & Propeller Configuration

- **Configuration**: We chose a single motor in a tractor (puller) configuration mounted on the nose. This arrangement places the motor mass forward, making it easier to align the center of gravity. While a dual-motor setup offers more thrust, the single motor reduces weight and eliminates the risk of asymmetric thrust due to motor misalignment.

- **Yaw Compensation**: The propeller direction is set counter-clockwise. This ensures that any yaw caused by asymmetric propeller loading acts in the direction the plane is turning around the pole (SKYbrary Aviation Safety, 2021).

- **Motor Mount**: The mount consists of a 3D-printed shell that fits over the fuselage nose and a mounting plate. It is secured via adhesive and a transverse bar clamping it through the foremost fuselage cutout. We 3D-printed it to ensure its strength and ability to withstand larger loads compared to the woods.

Selection and Use of Materials

- **Balsa Wood**: Used for the middle fuselage layer and the internal wing ribs. This maximises weight savings in non-load-bearing areas, directly contributing to a lower stall speed.

- **Plywood:** Utilised for the outer fuselage layers.This provides the necessary rigidity and toughness for the main body to handle the stresses involved during flight.

- **3D Printing (PLA)**: Used for complex, high-stress components like the wing root and wing tips. This allows for more precise aerodynamic design and impact resistance that laser cutting wood cannot easily provide.

- **Vinyl Film**: Chosen for the wing skin. It is lightweight and provides a smooth surface finish, reducing skin friction drag.

Parts List

- 4x Plastic Airfoils
- 12x Balsa Airfoils
- Vinyl Wing Skin
- 2x Carbon Wing Spar
- Balsa fuselage middle section
- Balsa horizontal stabiliser
- 2x Plywood fuselage
- Motor Mount
- Motor Mount Support Bar
- Wing Root
- Forward Landing Gear
- Rear Landing Skid
- 4x Bolts for Motor Mounting
- 6x Screws for Wing Mounting
- 10x Screws for Counterweight

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